

PROJECT 4

Enterprise Network Monitoring & NOC Incident Response Lab

Built with EVE-NG, Cisco IOS, Kali Linux, and Zabbix

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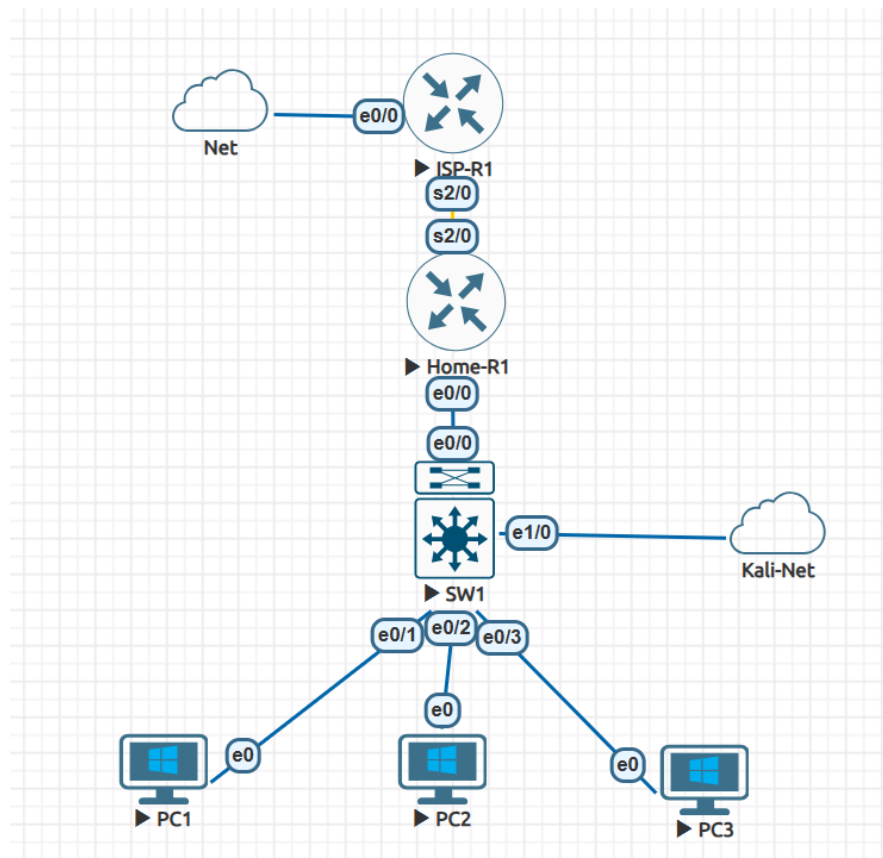


Figure 1: EVE-NG Lab Topology

1. Project Overview

This project is an end-to-end Network Operations Center (NOC) simulation built in EVE-NG. It combines Cisco IOS routers/switches, Kali Linux, and Zabbix to demonstrate infrastructure monitoring, NOC dashboarding, and structured incident response across three escalating failure scenarios: a single-host outage, a core gateway outage, and a WAN link outage.

Key objectives demonstrated:

- Infrastructure monitoring: host discovery, ICMP polling, and alert thresholds in Zabbix
- NOC dashboarding: real-time availability widgets for rapid visual triage
- Structured incident response lifecycle: Detection → Triage → Diagnostics → Root Cause Analysis → Remediation → Verification → Reporting
- Root cause analysis distinguishing single-host failures from core gateway / single-point-of-failure (SPOF) outages

2. Network Topology & Architecture

The topology consists of ISP-R1 (WAN edge) connected via a serial link to Home-R1, which provides the LAN gateway through SW1. SW1 connects PC1, PC2, and PC3, along with the Kali Linux Zabbix server on a separate Kali-Net segment (interface e1/0).



(See Figure 1 on the title page for the exact EVE-NG topology screenshot.)

3. IP Addressing Scheme

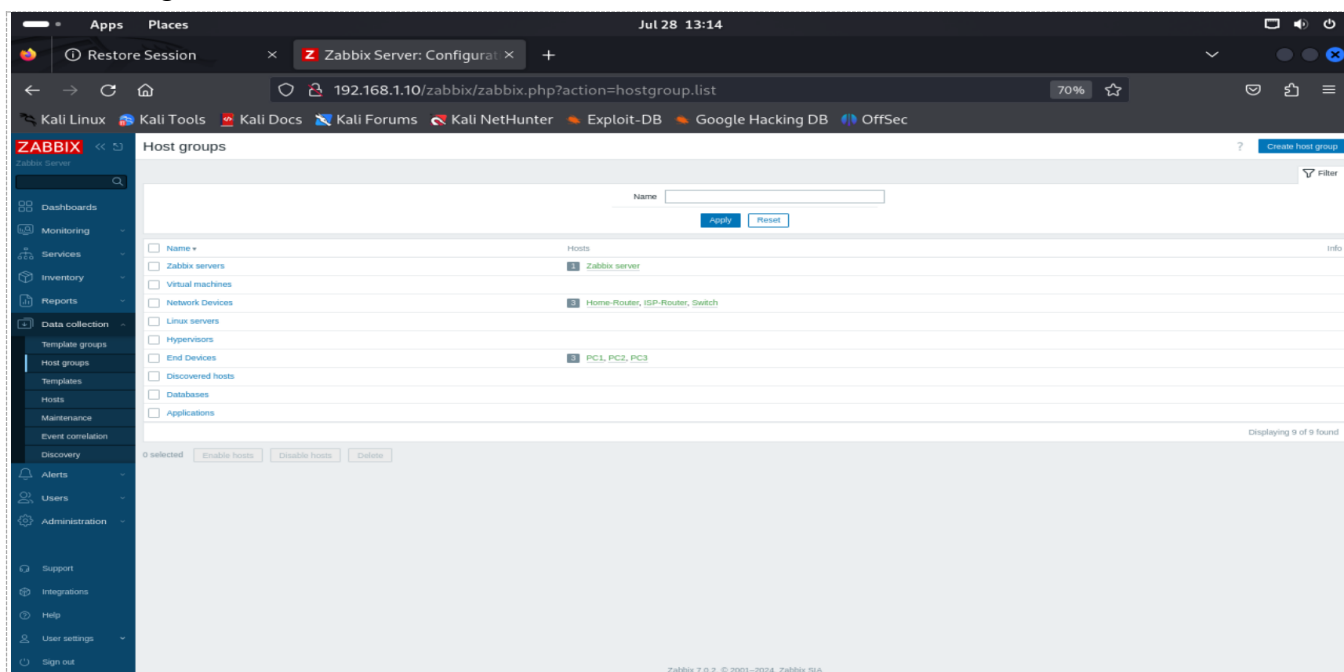
Device	Interface / Role	IP Address
ISP-R1	e0/0 (WAN, toward Net)	8.8.8.8/24
ISP-R1	s2/0 (link to Home-R1)	10.0.0.1/24
Home-R1	s2/0 (WAN, toward ISP-R1)	10.0.0.2/24
Home-R1	e0/0 (LAN, toward SW1)	192.168.1.1/24
Kali Linux / Zabbix	e1/0 (via SW1, Kali-Net)	192.168.1.10/24 (static)
SW1	Vlan1	192.168.1.2
PC1	e0 (via SW1, e0/1)	DHCP (192.168.1.1–200)

PC2	e0 (via SW1, e0/2)	DHCP (192.168.1.1–200)
PC3	e0 (via SW1, e0/3)	DHCP (192.168.1.1–200)

4. Zabbix Monitoring Configuration

Phase 1 Setup. Performed in the Zabbix Web UI (<http://192.168.1.10/zabbix>):

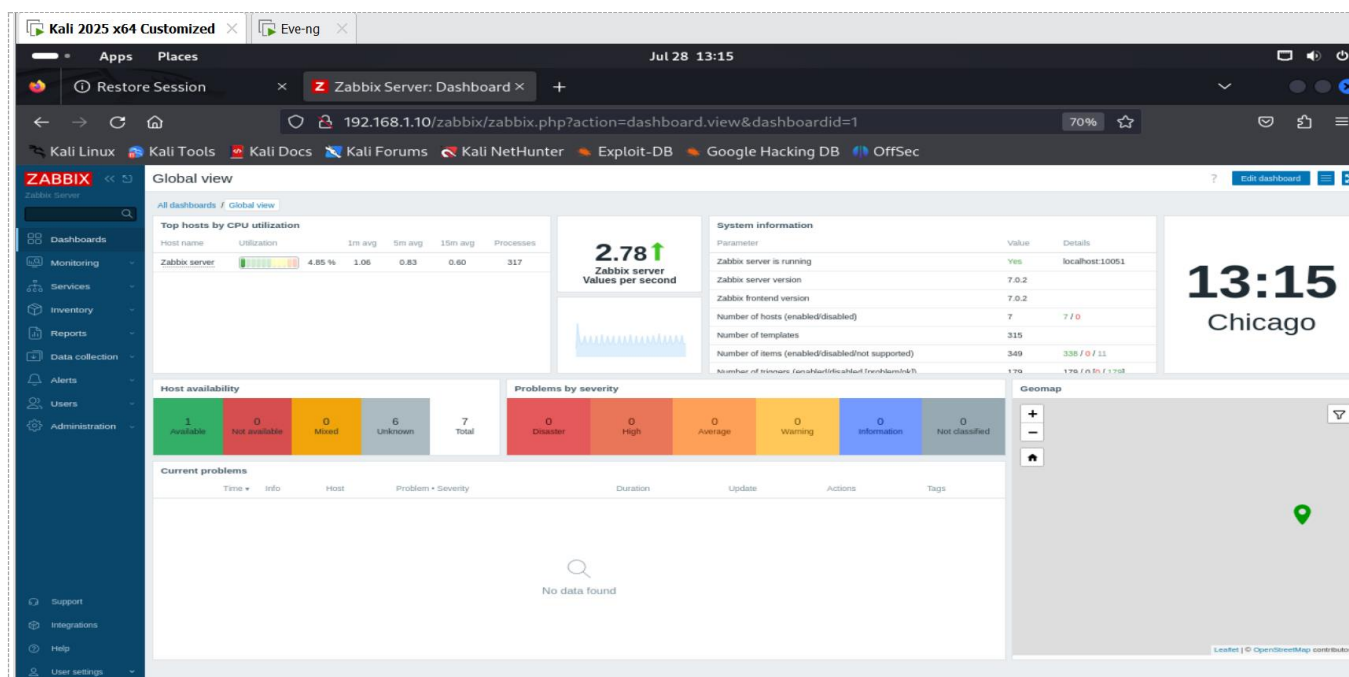
1. Created two host groups under Configuration → Host Groups: "Network Devices" and "End Devices."
2. Added hosts under Configuration → Hosts → Create Host: ISP-Router (10.0.0.1), Home-Router (192.168.1.1), Switch, and PC1–PC3.
3. Linked the standard ICMP Ping template (Template Module ICMP Ping) to all hosts for availability tracking.



5. NOC Dashboard & Baseline

Phase 2 Baseline. Built a dashboard to give at-a-glance visibility into the environment:

1. Created a new dashboard under Dashboards → Create Dashboard.
2. Added a Host Availability widget grouped by "Network Devices" and "End Devices."
3. Added a Problems widget filtered for severity Warning and higher.
4. Verified all status indicators showed UP before starting incident simulations.



6. Incident Simulations

Phase 3 Operations. Three incidents were simulated sequentially, each escalating in scope and severity: a single access-layer host failure, a core gateway failure affecting the whole LAN, and a WAN edge failure affecting only external reachability.

6.1 INC-001 Host Connectivity Failure

Incident ID	INC-001
Title	Host Connectivity Failure
Severity	Low / Medium
Target Device	PC2 (192.168.1.5)
Detection Method	Zabbix ICMP Ping Trigger
Detection Time	2026-07-28 11:29:42 PM
Symptom	Zabbix fired the trigger "ICMP ping loss - PC2 is unreachable." Impact was limited to a single host offline; no other devices affected.
Investigation	Ran an ICMP check from the Kali NOC workstation to PC2 and observed 100% packet loss. Checked the switch interface port status connecting PC2 and found the link down, consistent with a simulated physical disconnect or interface shutdown.
Root Cause	Physical or logical interface drop on the access-layer switch port connecting PC2.

Resolution	Re-enabled the switchport / reconnected the interface for PC2.
Verification	Confirmed ICMP reply restored from PC2. Zabbix status returned to OK and a recovery notification was logged.
Status	Resolved

The image displays two screenshots of the Zabbix 7.0.2 dashboard, showing the state of the monitoring system during an incident response.

Top Screenshot (Initial State):

- Global view:** Shows the Zabbix server's status and performance metrics.
- Top hosts by CPU utilization:** Zabbix server utilization is 1.41%.
- System information:** Zabbix server is running, version 7.0.2.
- Host availability:** 1 Available, 0 Not available, 0 Mixed, 6 Unknown, 7 Total.
- Problems by severity:** 0 Disaster, 1 High, 0 Average, 0 Warning, 0 Information, 0 Not classified.
- Current problems:** A problem is listed for PC2: "Unavailable by ICMP ping" (Severity: High, Duration: 1m 20s).
- Geomap:** Shows the location of the Zabbix server in Chicago.

Bottom Screenshot (Verification State):

- Global view:** Similar to the top screenshot, but the 'Zabbix server Values per second' graph shows a decrease to 2.71.
- Current problems:** The problem for PC2 is no longer listed, indicating it has been resolved.
- Terminal Window:** A terminal window is open, showing the results of a ping command to 192.168.1.5. The output indicates that the ping is successful, with 4 packets transmitted and 4 received, 0% packet loss, and a time of 3014ms.

6.2 INC-002 Core Gateway Outage

Incident ID	INC-002
Title	Core Gateway Outage
Severity	High / Critical
Target Device	Home-Router (192.168.1.1)
Detection Method	Zabbix ICMP Ping Trigger Group
Detection Time	2026-07-28 11:46:34 PM
Symptom	Triggers fired simultaneously across ISP-Router and Home-Router itself. Impact was complete LAN degradation with zero outbound traffic possible.
Investigation	As NOC analyst, evaluated the pattern of simultaneous host failures across the LAN. Identified Home-Router as the common gateway and single point of failure (SPOF) for all downstream hosts. Console connection to the router confirmed its power/process state was down.
Root Cause	Home-Router node was shut down / suspended, removing the default gateway for the entire LAN segment.
Resolution	Restarted the router process and re-enabled the core link (Home-Router node).
Verification	Verified gateway reachability with ping 192.168.1.1. All downstream hosts automatically cleared their alerts in Zabbix within 120 seconds.
Status	Resolved

The image shows a Kali Linux desktop environment with two browser windows open. The top window displays the Zabbix Server Dashboard (Global view) for the IP 192.168.1.10. The dashboard includes a sidebar with navigation options like Dashboards, Monitoring, Services, Inventory, Reports, Data collection, Alerts, Users, and Administration. The main content area shows 'Global view' with sections for 'Top hosts by CPU utilization', 'System information', 'Host availability', 'Problems by severity', and 'Current problems'. A large clock in the top right corner shows 13:50 in Chicago. The bottom window displays the 'Latest data' view for the same Zabbix server, showing a detailed table of monitored items and their values. The table includes columns for Host, Name, Last check, Last value, Change, Tags, and Info. The table lists various system metrics, network interface statistics, and service health checks.

Global view Dashboard Data:

- Top hosts by CPU utilization:**

Host name	Utilization	1m avg	5m avg	15m avg	Processes
Zabbix server	5.66%	0.27	0.39	0.45	315

- System information:**

Parameter	Value	Details
Zabbix server is running	Yes	localhost:10051
Zabbix server version	7.0.2	
Zabbix frontend version	7.0.2	
Number of hosts (enabled/disabled)	7	7 / 0
Number of templates	315	
Number of items (enabled/disabled/not supported)	349	338 / 0 / 11
Number of triggers (enabled/disabled/trouble/shutdown)	179	179 / 0 / 0

- Host availability:**

Available	Not available	Mixed	Unknown	Total
1	0	0	6	7

- Problems by severity:**

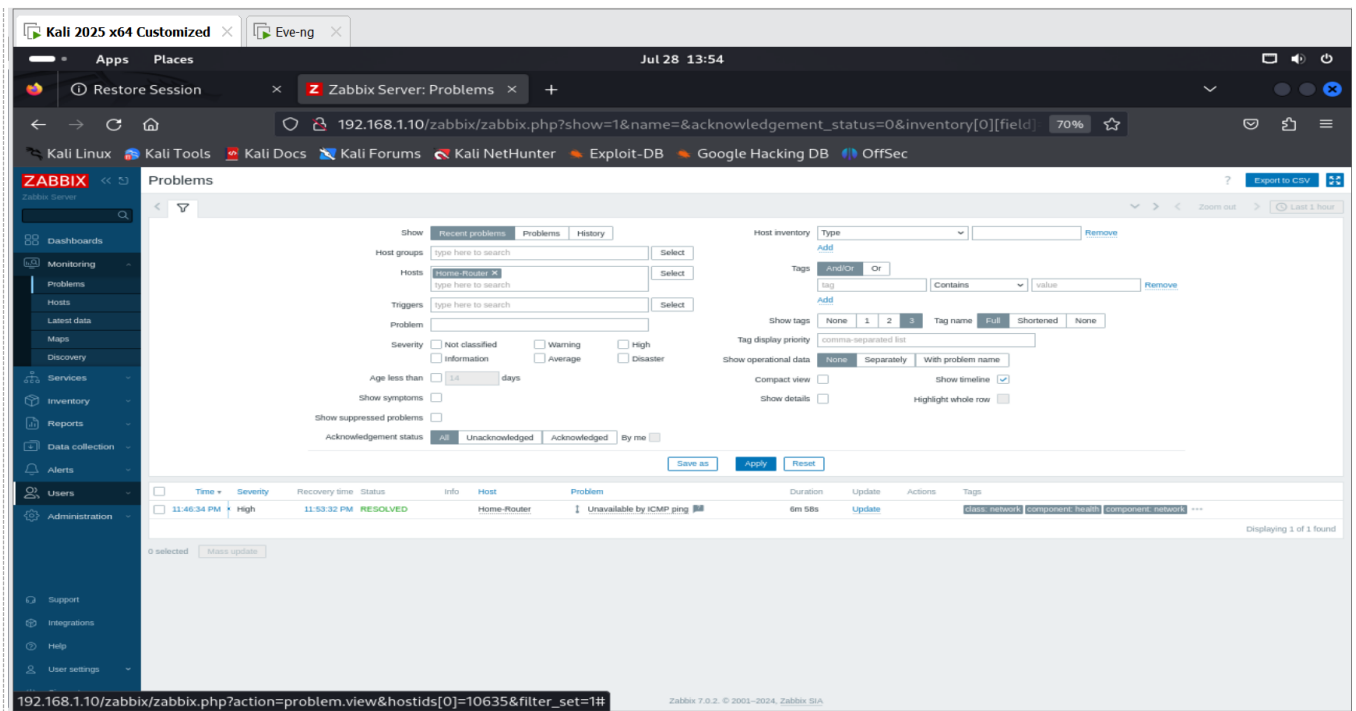
Disaster	High	Average	Warning	Information	Not classified
0	2	0	0	0	0

- Current problems:**

Time	Host	Problem + Severity	Duration	Update	Actions	Tags
11:46:34 PM	Home-Router	Unavailable by ICMP ping	4m 11s	Update		class: network, component: health, component: network
11:46:31 PM	ISP-Router	Unavailable by ICMP ping	4m 14s	Update		class: network, component: health, component: network

Latest data Table Data:

Host	Name	Last check	Last value	Change	Tags	Info
Home-Router	ICMP loss	21s	100 %		component: health, component: network	Graph
ISP-Router	ICMP loss	24s	100 %		component: health, component: network	Graph
Switch	ICMP loss	21s	0 %		component: health, component: network	Graph
Home-Router	ICMP ping	21s	Down (0)		component: health, component: network	Graph
ISP-Router	ICMP ping	24s	Down (0)		component: health, component: network	Graph
Switch	ICMP ping	21s	Up (1)		component: health, component: network	Graph
Home-Router	ICMP response time	21s	0		component: health, component: network	Graph
ISP-Router	ICMP response time	24s	0		component: health, component: network	Graph
Switch	ICMP response time	21s	1.26ms	-0.15ms	component: health, component: network	Graph
Switch	Interface E100/0: Bits received	19s	0 kbps		component: network, description: Interface: E100	Graph
Switch	Interface E100/0: Bits sent	19s	1.1 Kbps	+24 kbps	component: network, description: Interface: E100	Graph
Switch	Interface E100/0: Inbound packets discarded	19s	0		component: network, description: Interface: E100	Graph
Switch	Interface E100/0: Inbound packets with errors	19s	0		component: network, description: Interface: E100	Graph
Switch	Interface E100/0: Interface type	10h 41m 16s	ethernetCsmacd (0)		component: network, description: Interface: E100	Graph
Switch	Interface E100/0: Operational status	19s	up (1)		component: network, description: Interface: E100	Graph
Switch	Interface E100/0: Outbound packets discarded	19s	0		component: network, description: Interface: E100	Graph
Switch	Interface E100/0: Outbound packets with errors	19s	0		component: network, description: Interface: E100	Graph
Switch	Interface E100/0: Speed	1m 19s	10 Mbps		component: network, description: Interface: E100	Graph
Home-Router	Interface E100/LAN-to-Switch: Bits received	9m 21s	1.5 Kbps	+48 kbps	component: network, description: LAN-to-Switch, Interface: E100	Graph
Home-Router	Interface E100/LAN-to-Switch: Bits sent	9m 21s	616 kbps	+56 kbps	component: network, description: LAN-to-Switch, Interface: E100	Graph



6.3 INC-003 WAN Link Outage

Incident ID	INC-003
Title	WAN Link Outage
Severity	High
Target Link	10.0.0.1/30 (Home-R1 ↔ ISP-R1)
Detection Method	Zabbix ICMP / Edge Monitoring
Detection Time	2026-07-29 12:00:31 AM
Symptom	Zabbix reported ISP-R1 (10.0.0.1) as down, while local LAN devices (PC1–PC3, Home-Router) remained reachable and healthy. This pointed to a fault beyond the LAN, on the WAN edge.
Investigation	Ran a traceroute to 10.0.0.1 from Kali Linux; the trace reached Home-Router (192.168.1.1) and then timed out, indicating the WAN hop was the failure point. Logged into the Home-Router CLI and ran "show ip interface brief," which showed the WAN interface as up/down (line protocol down).
Root Cause	WAN link failure between Home-Router and ISP-R1 serial/Ethernet interface link down, simulating an upstream provider or physical WAN port failure.
Resolution	Restored the WAN serial/Ethernet connection between Home-Router and ISP-R1.

Verification	Verified routing table convergence with "show ip route." Zabbix confirmed the WAN edge host (ISP-R1) was reachable again.
Status	Resolved

```

ISP-R1 Home-R1 SW1
*Jul 28 18:58:11.187: %CDP-4-DUPLEX_MISMATCH: duplex mismatch discovered on Ethernet0/0 (not half duplex), with Switch.lab.local Ethernet0/0 (half duplex).
Home-Router(config-if)#
*Jul 28 18:59:01.998: %CDP-4-DUPLEX_MISMATCH: duplex mismatch discovered on Ethernet0/0 (not half duplex), with Switch.lab.local Ethernet0/0 (half duplex).
Home-Router(config-if)#
*Jul 28 18:59:57.904: %CDP-4-DUPLEX_MISMATCH: duplex mismatch discovered on Ethernet0/0 (not half duplex), with Switch.lab.local Ethernet0/0 (half duplex).
Home-Router(config-if)#
*Jul 28 19:00:52.090: %CDP-4-DUPLEX_MISMATCH: duplex mismatch discovered on Ethernet0/0 (not half duplex), with Switch.lab.local Ethernet0/0 (half duplex).
Home-Router(config-if)#
*Jul 28 19:01:40.107: %CDP-4-DUPLEX_MISMATCH: duplex mismatch discovered on Ethernet0/0 (not half duplex), with Switch.lab.local Ethernet0/0 (half duplex).
Home-Router(config-if)#do sh ip int br
Interface IP-Address OK? Method Status Protocol
Ethernet0/0 192.168.1.1 YES NVRAM up up
Ethernet0/1 unassigned YES NVRAM administratively down down
Ethernet0/2 unassigned YES NVRAM administratively down down
Ethernet0/3 unassigned YES NVRAM administratively down down
Ethernet1/0 unassigned YES NVRAM administratively down down
Ethernet1/1 unassigned YES NVRAM administratively down down
Ethernet1/2 unassigned YES NVRAM administratively down down
Ethernet1/3 unassigned YES NVRAM administratively down down
Serial2/0 10.0.0.2 YES NVRAM administratively down down
Serial2/1 unassigned YES NVRAM administratively down down
Serial2/2 unassigned YES NVRAM administratively down down
--More--

kali@kali: ~
(kali@kali)-[~]
$ ping -c 4 192.168.1.5
PING 192.168.1.5 (192.168.1.5) 56(84) bytes of data.
64 bytes from 192.168.1.5: icmp_seq=1 ttl=128 time=1.74 ms
64 bytes from 192.168.1.5: icmp_seq=2 ttl=128 time=2.75 ms
64 bytes from 192.168.1.5: icmp_seq=3 ttl=128 time=2.93 ms
64 bytes from 192.168.1.5: icmp_seq=4 ttl=128 time=2.52 ms

--- 192.168.1.5 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3018ms
rtt min/avg/max/mdev = 1.735/2.482/2.926/0.455 ms

(kali@kali)-[~]
$ traceroute 10.0.0.1
traceroute to 10.0.0.1 (10.0.0.1), 30 hops max, 60 byte packets
 1 _gateway (192.168.1.1) 1.754 ms 1.725 ms 1.661 ms
 2 * * *
 3 * * *
 4 * * *
 5 * * *
 6 *^C

(kali@kali)-[~]
$

```

The screenshot shows the Zabbix 7.0.2 web interface in a browser window. The URL is `192.168.1.10/zabbix/zabbix.php?show=1&name=&acknowledgement_status=0&inventory[0][field]=70%`. The left sidebar contains navigation links for Dashboards, Monitoring, Problems, Hosts, Latest data, Maps, Discovery, Services, Inventory, Reports, Data collection, Alerts, Users, and Administration. The main content area is titled 'Problems' and includes filters for Host groups, Hosts, Triggers, Problem, Severity, Age less than, Show symptoms, Show suppressed problems, and Acknowledgement status. A table displays the problem details:

Time	Severity	Recovery time	Status	Info	Host	Problem	Duration	Update	Actions	Tags
12:00:31 AM	High	12:03:31 AM	RESOLVED		ISP-Router	Unavailable by ICMP ping	3m	Update		class: network, component: health, component: network

The bottom status bar shows the URL `192.168.1.10/zabbix/zabbix.php?action=problem.view&hostids[0]=10634&filter_set=1#` and the Zabbix version `Zabbix 7.0.2 © 2001-2024, Zabbix SIA`.

7. Conclusion

This lab demonstrates a realistic NOC workflow across three distinct failure classes: an isolated endpoint outage, a core gateway/SPOF failure, and a WAN edge failure. In each case, the incident was detected by Zabbix, triaged and diagnosed using CLI tools (ping, traceroute, show ip interface brief, show ip route), resolved, and formally verified and documented mirroring the detection-to-resolution lifecycle expected of a NOC or SOC analyst in a production environment.